Joha Park, Ph.D.

email: johapark@gmail.com
webpage: johapark.github.io

Computational biologist with 10+ years of experience spanning RNA biology, genomics, spatial omics, and CRISPR-Cas9 gene editing. Proven track record building AI/ML frameworks and scalable bioinformatics pipelines that translate large-scale sequencing and imaging data into therapeutic insights. Author of 8 high-impact publications. Currently driving human population-aware CRISPR off-target discovery, protein and RNA engineering, and variant effect prediction for genotoxicity derisking. Building agentic AI for research paper ingest

CURRENT POSITION

07/2026 – present **Principal Computational Scientist** *Intellia Therapeutics*
03/2023 – 06/2026 Senior Computational Scientist
>3 yrs Computational Sciences
Genomic Innovation
Intellia Therapeutics, Inc., Cambridge, MA, US

Roles

- Develop graph-based genomics methods integrating human genetic variation (e.g. gnomAD, 1KG, etc.) for population-aware CRISPR off-target prediction
- Apply genomic foundation models to predict variant effects and assess genotoxicity risks
- Develop deep learning models to predict CRISPR-Cas9 on- and off-target activities
- Build scalable, reproducible analysis pipelines for large-scale genomic and multi-omics
- Apply generative AI to protein and guide RNA engineering for therapeutic genome editing

PROFESSIONAL EXPERIENCE

01/2020 – 03/2023 **Postdoctoral Associate/Fellow** *Massachusetts Institute of Technology*
3 yrs 3 mos The Picower Institute for Learning and Memory
Institute for Medical Engineering and Science
Department of Chemical Engineering
Massachusetts Institute of Technology, Cambridge, MA, US

Roles

- Develop hydrogel-based tissue clearing and expansion methods for spatial omics
- Build a scalable image processing AI/ML framework for 3D light-sheet microscopy
- Generate a 3D single-cell mouse brain reference atlas

03/2019 – 12/2019 **Postdoctoral Researcher** *Seoul National University*
10 months Center for RNA Research
Institute for Basic Science
Seoul National University, Seoul, Korea

Roles

- Lead projects on mRNA poly(A)-tail length regulations
- Conduct both wet and dry lab experiments for RNA biology
- Develop and analyze custom NGS applications

EDUCATION AND TRAINING

2012 – 2019	**Ph.D. in Biological Sciences** (RNA Biology & Bioinformatics) RNA Biology Lab (Advisor: V. Narry Kim) Dissertation: “Poly(A) length regulation: deadenylases and the poly(A) barricade”	*Seoul National University*
2015 – 2018	**Expert Research Personnel** (Compulsory Military Service)	*Republic of Korea Army*
2008 – 2012	**B.S. in Biological Sciences** (*cum laude*)	*Seoul National University*

PUBLICATIONS

*: co-first author, #: co-corresponding author

1 **J. Park***, M. Kim*, H. Yi*, K. Baeg*, Y. Choi, Y.-s. Lee, J. Lim, V. N. Kim (2023) “Short poly(A) tails are protected from deadenylation by the LARP1-PABP complex” ***Nat. Struct. Mol. Biol.,*** 30(3):330–338

2 **J. Park***, S. Khan*, D. H. Yun*, T. Ku, K. L. Villa, J. E. Lee, Q. Zhang, G. Feng, J. Park, E. Nedivi#, K. Chung# (2021) “Epitope-preserving magnified analysis of proteome (eMAP)” ***Sci. Adv.*** 7(16):eabf6589

3 Y. Kim*#, **J. Park***, S. Kim*, M. Kim, M.-G. Kang, C. Kwak, M. Kang, B. Kim, H.-W. Rhee, and V. N. Kim# (2018) “PKR senses nuclear and mitochondrial signals by interacting with endogenous double-stranded RNAs” ***Mol. Cell***, 71(6):1051–1063.e6
- Selected as “Research Highlights” in ***Nat. Chem. Biol.***, 14(11):989

4 H. Yi*, **J. Park***, M. Ha, J. Lim, H. Chang and V. N. Kim (2018) “PABP cooperates with the CCR4-NOT complex to promote mRNA deadenylation and block precocious decay” ***Mol. Cell***, 70(6):1081–1088.e5
- Previewed in ***Mol. Cell***, 70(6):987–988

5 T. A. Nguyen*#, **J. Park***, T. L. Dang, Y.-G Choi, and V. N. Kim# (2018) “Microprocessor depends on hemin to recognize the apical loop of primary microRNA” ***Nucleic Acids Res.***, 46(11):5726–5736

6 K. T. You, **J. Park**, and V. N. Kim (2015) "Role of the small subunit processome in the maintenance of pluripotent stem cells" ***Genes Dev.***, 29(19):2004–2009

7 T. A. Nguyen, M. H. Jo, Y.-G. Choi, **J. Park**, S. C. Kwon, S. Hohng, V. N. Kim# and J.-S. Woo# (2015) “Functional anatomy of the human Microprocessor” ***Cell***, 161(6):1374–1387

8 Y.-K. Kim, G. Wee, **J. Park**, J. Kim, D. Baek, J.-S. Kim#, and V. N. Kim# (2013) “TALEN-based knockout library for human microRNAs” ***Nat. Struct. Mol. Biol.***, 20(12):1458–1464

Links: Google Scholar, PubMed

HONORS AND AWARDS

2022 – 2023 Picower Postdoctoral Fellows, *The Picower Institute for Learning and Memory, MIT*

2021 AKN Outstanding Research Awards, *Association of Korean Neuroscientists*

2019 Conference Scholarship, *Keystone Symposia: Small Regulatory RNAs*

2019 Best Dissertation Awards, *School of Biological Sciences, SNU*

2008 – 2012 Presidential Science Scholarship, *Korea Science and Engineering Foundation*

INTERNATIONAL CONFERENCES

2019 Cell Symposia: Regulatory RNAs — ***short talk***
"Poly(A) barricade as a rate-limiting factor that shapes poly(A) tail length"

2018 CSHL Meeting: Regulatory & Noncoding RNAs — ***short talk***
"PKR senses nuclear and mitochondrial signals by interacting with endogenous dsRNAs"

2017 CSHL Meeting: Eukaryotic mRNA Processing — ***short talk***
"Genome-wide evaluation of the role and specificity of deadenylases"

2016 EMBL Symposium: The Complex Life of mRNA — *poster*

2016 IMBA: 11th Microsymposium on Small RNAs — *poster*

2014 Keystone Symposia: RNA Silencing — *poster*

INVITED TALKS

08/2023 Special Seminar – *Graduate School of Medical Science and Engineering, KAIST*

11/2022 Special Seminar – *Yonsei University College of Medicine*

08/2022 Young Global Leader Symposium – *Dept. of Chemical and Biomolecular Engineering, KAIST*

07/2022 Department Seminar – *Department of Pharmacology, SNU College of Medicine*

05/2022 Young Investigator Seminar Series – *Department of Biomedical Engineering, UNIST*

TEACHING EXPERIENCE

2022 Kaufman Teaching Certificate Program, *Massachusetts Institute of Technology*

2014 Teaching Assistant, Advanced RNA Biology Lab, *School of Biological Sciences, SNU*

2013 Teaching Assistant, Biology Lab 2, *School of Biological Sciences, SNU*

2012 Teaching Assistant, Biology Lab 1, *School of Biological Sciences, SNU*

2011 Tutor, Basic Course for Biology, *School of Biological Sciences, SNU*

TECHNICAL EXPERTISE

Computational biology

- AI/ML
  - Biological LLMs: variant effect prediction, protein/RNA sequence embeddings
  - Model training and fine-tuning: PyTorch, Scikit-learn
  - Binding affinity prediction, genotoxicity risk scoring
  - Generative AI for protein engineering: BoltzGen, ProteinMPNN, etc.
  - Agentic AI for personal knowledge base: LLM-based paper ingest and cross-corpus literature synthesis for scientific research (GitHub)

- Next-generation sequencing (NGS)
  - (sc)RNA-seq, WGS, WES, ATAC-seq, Amp-seq, CLIP-seq, TAIL-seq, etc.
  - Statistical analysis for genomics and transcriptomics
  - Tools: Samtools, bcftools, GATK, Scanpy, Seurat, STAR, Bedtools, etc.
- Image processing
  - Concurrent programming for large-scale images
  - 2D/3D rigid and non-rigid biomedical image registration
  - Convolutional Neural Networks
  - Tools: PyTorch, Zarr, Dask, Scikit-image, OpenCV, ImageJ/Fiji, Imaris, etc.
- General skills
  - Unix/Linux, AWS cloud computing environment
  - Programming languages: Python, Rust, R
  - Python scientific computing libraries: NumPy, SciPy, Pandas, etc.
  - Python machine learning frameworks: Scikit-learn, PyTorch, Tensorflow
  - Building a reproducible analysis pipeline: Nextflow, Snakemake, Docker, Singularity
  - Efficient data visualization: Matplotlib, Seaborn, Plotly

Wet experiments

- Routine experiments
  - Molecular cloning: Gibson, Golden Gate assembly
  - Mammalian, bacterial cell culture
  - DNA and siRNA transfection
  - Quantitative RT-PCR
  - Immunoprecipitation
  - Western blotting
- RNA biology
  - Custom sequencing library preparation
  - Custom high-resolution poly(A)-tail length assay
  - Tet-On inducible reporter cell line generation
  - RNA co-immunoprecipitation
  - Lentiviral transduction
  - *In vitro* transcription
  - Northern blotting
  - Luciferase reporter assay
- Tissue clearing and expansion
  - SHIELD clearing
  - Expansion microscopy: eMAP